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Introduction

Online shopping is becoming more and more popular in recent years. Many products that consumers intend to buy may not be available at their nearby store. Online Shopping provides the ability to purchase any necessity from the comfort of your home, saving time by avoiding crowded stores and traffic, provides the ability to compare the prices of products across stores, and provides a wide selection of products. Online shopping eliminates the need to visit various locations and is globally accessible. Online shopping is only becoming more popular, so much so that having an online presence is a necessity for a business to be competitive in their field and is one of the many methods businesses seek to increase their revenue.

This project focuses on the online aspect of consumers. Using the combination of a predictive and clustering model, we will construct a decision matrix utilizing both models to have a tailored decision for every customer group of customers. Identifying which type of consumers can be targeted with an ad or not will increase the profit of any given business. By targeting the right type of consumer who is more likely to be persuaded into purchasing an item, we will minimize meaningless losses and expenses while maximizing the possible return of ads, promotions, and suggestions.

Methods

We use the Online Shoppers Purchasing Intention Dataset from UC Irvine. Among the 12,330 online shopping sessions, 84.5% (10,442) have a negative target value, indicating that the session ended without a purchase, while the remaining 15.5% (1,908) resulted in a positive target value.

The dataset includes 17 features with a binary target variable, Revenue, reflecting whether or not a user completed a purchase. Three sets of page-browsing behavioral features provided, Administrative, Informational, and ProductRelated, along with their respective duration variables. These features describe both the number of pages visited and the total time spent within each category. As shown in the correlation feature matrix below, variables within each group exhibit strong internal correlations, and the groups themselves are positively correlated.

Additional behavioral indicators include Bounce Rate, the percentage of visitors who land on a page and leave without further interaction; Exit Rate, the percentage of sessions that end on a given page; and Page Value, the average value of a page based on the likelihood of it appearing prior to a purchase. These variables collectively capture key aspects of user engagement and browsing behavior within an online shopping session.

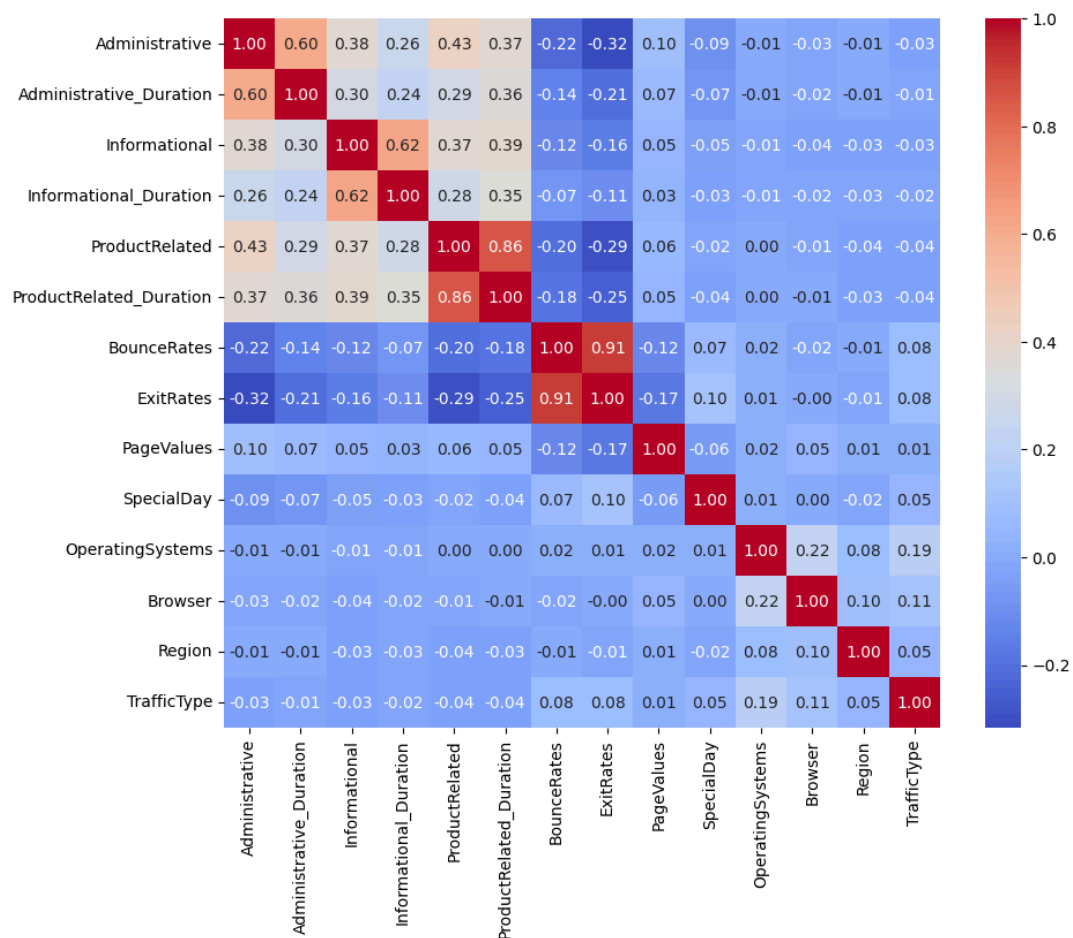


Figure 1: Correlation matrix of data

Our proposed decision system consists of two components, a predictive model and a clustering model. The predictive model estimates user purchase intention based on their attributes, while the clustering model identifies groups of users who share similar behavioral patterns reflected in those same attributes. Because both models are trained on the same dataset with the same features, a high degree of overlap between the resulting classes is expected. For instance, users who leave the website immediately tend to have very low purchase intention, whereas those who browse for a longer period are more likely to make a purchase. We combine the outputs from both models to construct a decision matrix, allowing us to leverage the data effectively and generate interpretable results.

Our predictive model aims to predict user's purchase intent, enabling personalized advertisements and promotions. 20 percent of data was set aside for testing, while the rest 80 percent of data was used for training. Logistic regression was used as a baseline, while random forest served as the primary predictive model. Both models were selected because their complexity aligns with our dataset size and task, and they are widely recognized approaches. After evaluating the models using weighted-f1 score, confusion matrix and ROC/AUC, we plotted the calibration plot of the prediction results.

The goal of the clustering model is to represent the behavioral engagement of each consumer and designate them into groups. The 14 variables selected were all related to the consumer's session behavior that include:

- Administrative
- Administrative_duration
- Informational
- Informational_Duration
- ProductRelated
- ProductRelated_Duration
- BounceRates
- ExitRates
- PageValues
- SpecialDay
- OperatingSystem
- Brower
- Region
- TrafficType

To form our clusters/groups, K-means was chosen as the method to form our clusters. K-means is great for large datasets. To determine the optimal number of

clusters, the elbow method was used. At first glance at the elbow method plot, there is not a clear “elbow point”. At no point in the plot does it plateau. The silhouette score is added to help evaluate the quality of a clustering as well as how well it separates and how well formed the clusters are. The higher the silhouette score the better and in our case $k=3$ has the highest silhouette score at ~ 0.32 . A silhouette score of 0.32 is considered low, but when it comes to real world customer behaviors where user behavior heavily overlaps and consumers do not fall into clean, clear groups, a score between 0.2-0.4 is considered acceptable. People online are unpredictable and there is no real “clear” distinction and boundary between each cluster of customers.

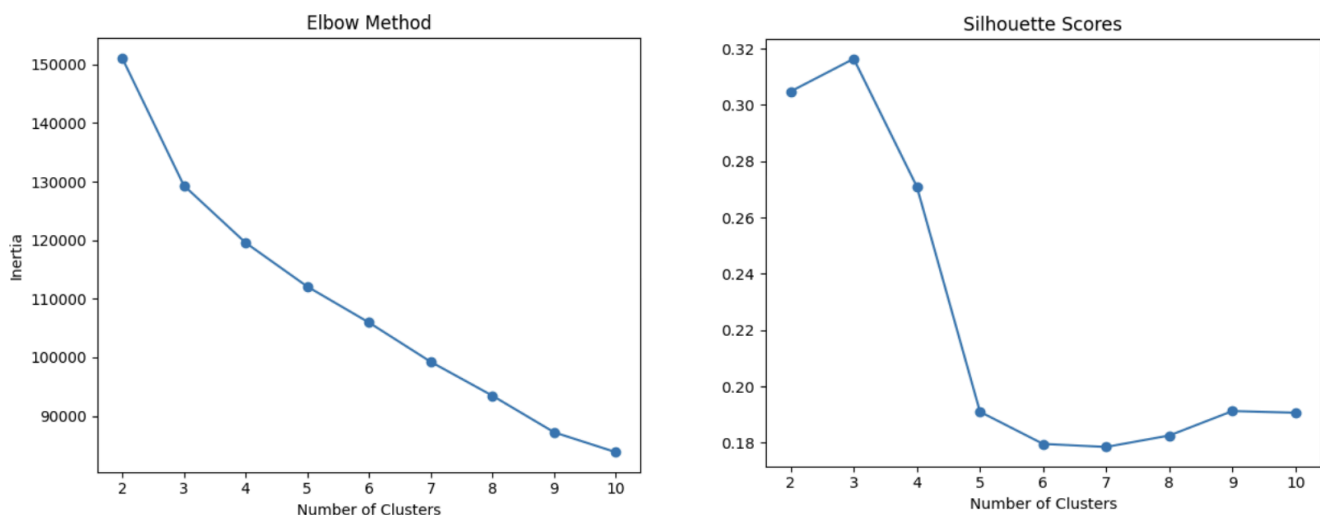


Figure 2 Finding optimal K-value

Results

The random forest model achieved a weighted F1 score of 0.89, compared to 0.85 for logistic regression, and demonstrated superior AUC. With a default threshold of 0.5, random forest shows higher precision for positive predictions, with a slightly lower recall. Given the importance of both error types in our task, we selected random forest as the predictive model.

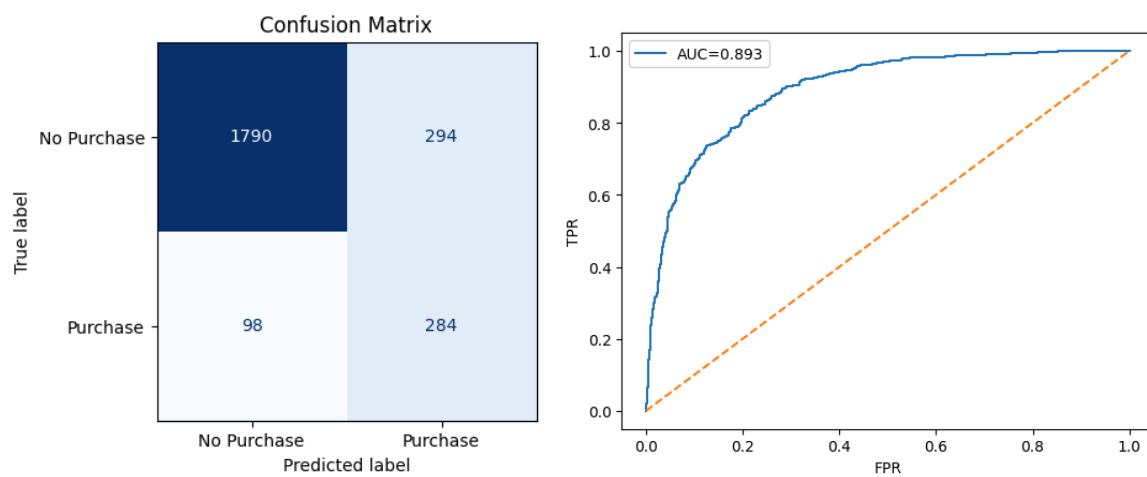


Figure 3 Logistic regression results

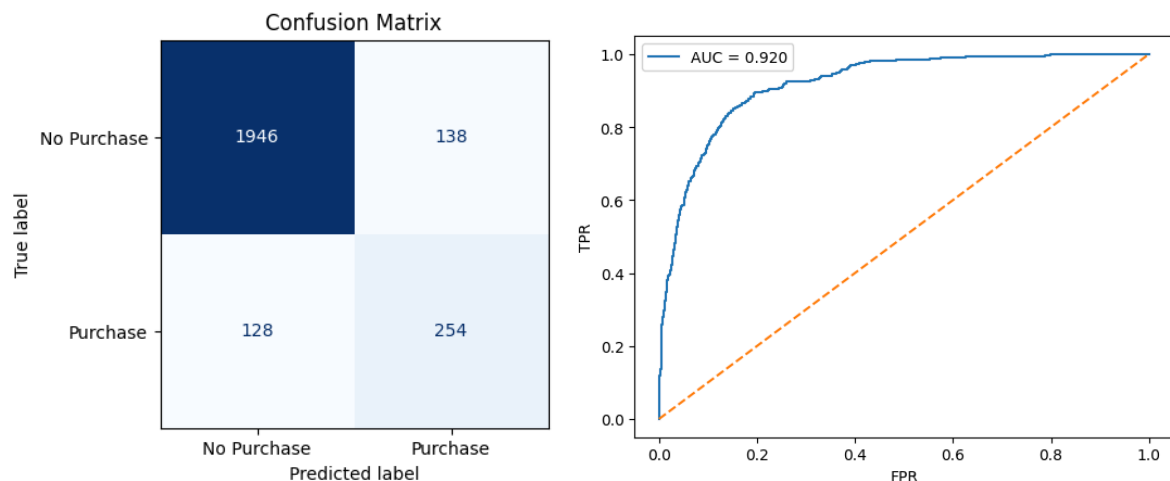


Figure 4 Random forest results

To interpret the prediction result of the random forest model, we generated a calibration plot(Figure 5) comparing predicted purchase rates against actual purchase

rates. We divided the test set into 20 equal-sized bins based on predicted scores and plotted each bin against the actual purchase rate within that bin. The clustering of bins in the bottom-left corner clearly reflects the class imbalance within the dataset. The 0.2 predicted score marks the point where the curve switches from flat to nearly linear, indicating that users beyond this point are more reliably ranked by the model. Another observation is that above 0.6, the actual rate aligns more closely with the predicted rate, suggesting that the model achieves higher precision above this threshold. Overall, the calibration plot demonstrates that the model predicts purchase rates well at both low and high intent levels.

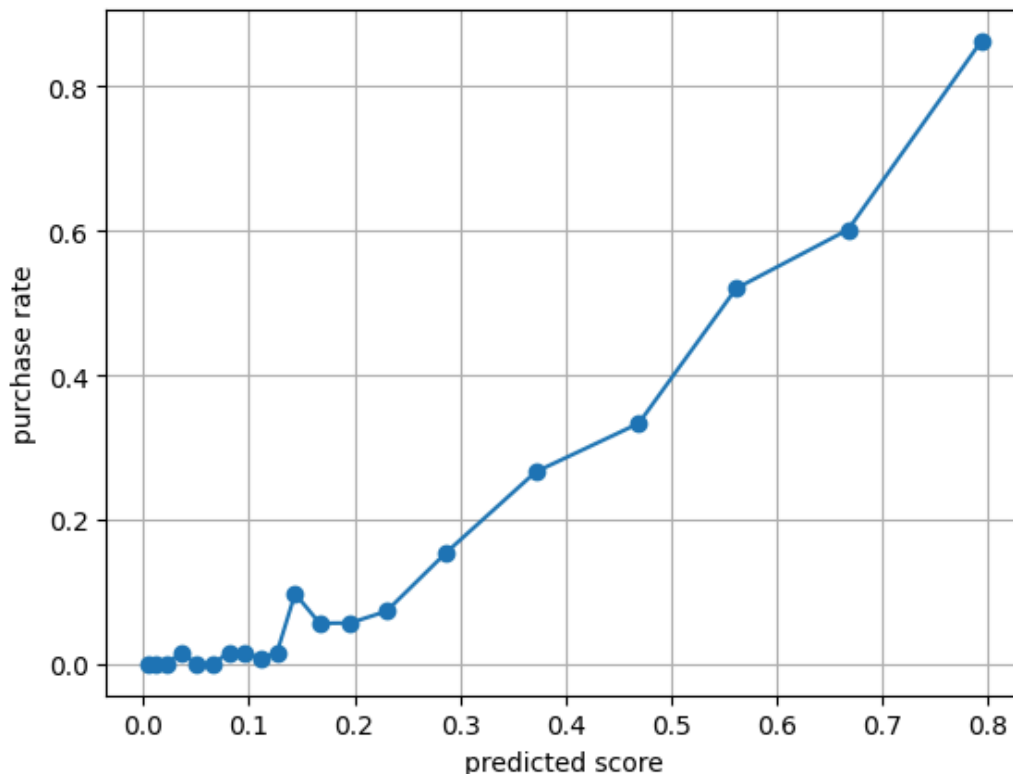


Figure 5 Calibration plot of random forest model

For our clustering model, our data was split into 3 clusters. Cluster 0 had the majority of the data with a size of 9,653. Cluster 1 was the second largest with the size of 1,622. And cluster 0 was the smallest with a cluster size of 1,055. The purchase rate of consumers in cluster 0 was 0.1503, the purchase rate of consumers in cluster 1 was 0.2780, and the purchase rate of consumers in cluster 2 was 0.0056. It can be seen that the majority of the data is clustered and heavily overlapping around the bottom left of the plot, reflecting the low, but acceptable silhouette score. Without the given colors to distinguish each cluster, it is shown that there is no clear boundary for each of the 3 clusters.

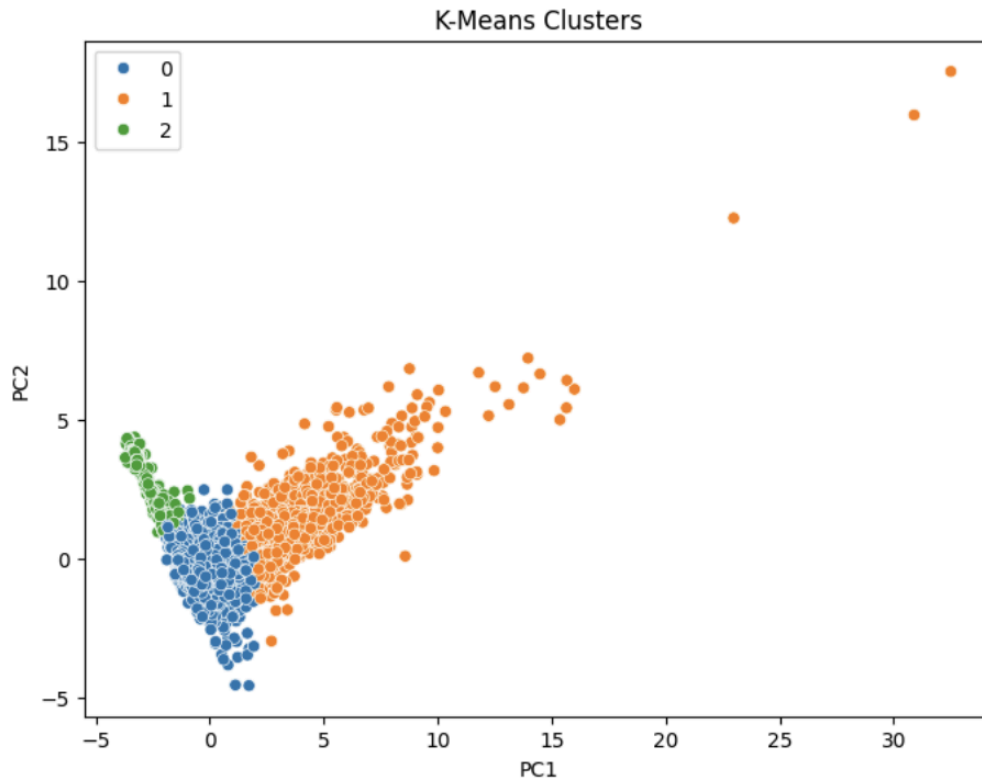


Figure 6 Clustering visualization

Discussion

Our clustering model already divides users into three groups. Those categorized in Cluster 0 are your average consumers and browser. These consumers all have a moderate number of pages visited, have a moderate amount of time spent on the products page, and have a moderate bounce/exit rate with a purchase likelihood rate of 15.03%. This is also where the bulk of the customers are categorized as. Those categorized in Cluster 1 are consumers who are serious shoppers and have a higher likelihood of purchasing a product. These consumers all have long session durations, a lot of page views, a low bounce/exit rate with a higher purchase likelihood rate of 27.8%. Those categorized in Cluster 2 are uninterested visitors, accidental clicks, bots, or even window shoppers. These visitors have a short session duration, almost zero page views, a high bounce/exit rate with a purchase likelihood of 0.56%. For a business, a high number of consumers categorized in Cluster 2 may indicate that the user is not provided with a good experience, the website is unpleasant to the eye, or may simply be the result of an accidental click. These results can not only be used to increase a businesses revenue, but also improve the business itself.

We decided to also split the results of our predictive model into 3 sections. Those with a prediction <0.2 are categorized as having a low purchase intention. Those with a model prediction from the ranges 0.2 to 0.6 are considered to have a moderate purchase intention. Lastly, those with a model prediction >0.6 are considered to have a high purchase intention.

The clustering model groups consumers based on browsing behavior and the predictive model estimates the purchase intent of consumers. Using the results from our prediction and behavioral clustering models, we will construct a decision matrix that utilizes both models to have a tailored decision for each unique group of customers.

Decision Matrix

	Low Model Prediction (<0.2)	Average Model Prediction ($0.2-0.6$)	High Model Prediction (>0.6)
Cluster 0 (Average user engagement)	Case 1	Case 2	Case 3
Cluster 1(High user engagement)	Case 4	Case 5	Case 6
Cluster 2(Low user engagement/ high exit rate)	Case 7	Case 8	Case 9

The case outcomes highlighted in green are interpreted as the situation where sending an advertisement, promotion, or a suggestion to a consumer will likely convert to an actual purchase. Case 2 has the consumers who are the most likely to be persuaded. Spending a budget on advertising, promoting, and suggesting products is likely to yield profit. For example, if a customer has been wanting a new water bottle, but was indecisive when purchasing the product due to the price, but the product is suddenly undergoing a 30% off sale on BlackFriday, the likelihood of that customer purchasing a new water bottle increases. Case 2 is where the majority of the consumers will be categorized and is where converting non-buyers to buyers will yield the bulk of the increased revenue. Case 8 is an interesting and rare situation. Those in case 8 have low user engagement but moderate model prediction. This group of consumers may not have a high browsing time, that does not mean they have no interest. An example of this type of behavior could be from concert-goers. Typically concert-goers purchase their ticket as soon as possible and exit the site as soon as possible. The decision action for case 8 may vary from business as some businesses may see less value in this grouping of customers.

The case outcomes highlighted in red are interpreted as the situation where a business should not invest budget sending advertisements, promotions, and suggestions to the customer. There is a very low likelihood the investment will convert to

an actual purchase. Sending an advertisement will not keep the online consumer on the page for much longer and is simply a waste of resources. Not investing in such consumers will help minimize business loss and help maximize yearly revenue.

The case outcomes highlighted in yellow is neither the case to send expensive advertisements nor the case to not send any advertisements at all. Case 3 and 6 includes consumers with moderate/high user engagement with high purchase intent. There is no need to spend advertisement budget on the customer, rather trying to upsell a product would be ideal in this situation. Upselling an item will not dissuade the customer from purchasing their product as they already have a positive mindset to purchase an item. Upselling products could potentially lead to an additional purchase thus increasing the business overall revenue.

Important Cluster + Predictive Model Interpretation

Case 1: Cluster 0(Average user engagement) + Low Model Prediction (<0.2)	Do not send Ads/Promotions -Uncertain, weak shopping intent and average behavior -casual browsing, accidental visits -ads and promotions are unlikely to change consumer behavior towards purchasing an item
Case 2: Cluster 0(Average user engagement) + Average Model Prediction (0.2-0.6)	Send ads/Promotions -There is where the bulk of the increased revenue comes from -Spending budget on these consumers is likely to yield profit -This group does not show the strongest purchase intent, but are persuadable (Convert non-buyers to buyers) -This is the group of consumers most influenced by ads/promotions Ex. Customers on black friday see a product they have been wanting for the longest time at 30% off.
Case 3: Cluster 0(Average user engagement) + High Model Prediction (>0.6)	Don't send Ads/Promotions, try upselling items -Customer already has high intention on buying an item -No need to waste spending ad budget on the consumer -Can yield profit through suggestion and upselling
Case 4: Cluster 1(High user engagement) + Low Model Prediction (<0.2)	Do Not Send Costly Ads/Promotions -Most interesting case: high engagement, long sessions but low purchase intent -rare case -comparison shoppers, high browsing without intent, research visitors, planning to buy later consumers, gathering information, undecided buyers, etc.
Case 5: Cluster 1(High user	Send Ads/Promotions, Recommendations, or Recommendations

engagement) + Average Model Prediction (0.2-0.6)	-potential buyer, their behavior is there, but unsure about their intent -the customer seems to think deeply about their decision before buying, waits for the best time to purchase Ex. Someone comparing the prices between 3 brands of computers for 30 minutes
Case 6: Cluster 1(High user engagement) + High Model Prediction (>0.6)	Do Not Send Ads/Promotions, try to show upsells/premium items -Consumer has high intention to buy item with high behavioral engagement -Save spending budget -Upselling will not disway customer as they already have a positive intent to purchase an item, potentially leading to an additional purchase
Case 7: Cluster 2(Low user engagement/ high exit rate) + Low Model Prediction (<0.2)	Do Not Send Costly Ads/Promotions, send small ads at most -Consumer is unlikely to buy anything -Expensive ads are a waste of resources and money for this group -Costumers might have clicked into the page by mistake, uninterested, frustrated by the site experience
Case 8: Cluster 2(Low user engagement/ high exit rate) + Average Model Prediction (0.2-0.6)	Send Ads/Promotions -rare and interesting case -low engagement and quick exits, but moderate intent -they don't browse a lot, but does not mean lack of interest -likely a combination of fast shoppers and knowing what they intend to purchase -possibly impulsive buyers or repeat purchases Ex. Possibly a case of a limited/seasonal product(concert tickets)
Case 9: Cluster 2(Low user engagement/ high exit rate) + High Model Prediction (>0.6)	Do Not Send Ads/Promotions -rare and interesting case -Likely a result of model error, contradictory actions, a really fast shopper -Ads/promotions unlikely impact the customer decision nor will it keep the customer on the page for much longer -

Conclusion

This project aimed to increase the online revenue of businesses by identifying customer behaviors and predicting their purchase intent through a random forest and k-means clustering model. We first ran a random forest model to predict the purchase

intent of a given customer, achieving an AUC of 0.920 and categorizing purchase intention into low, moderate, and high probability groups. Then we ran a k-means clustering model in order to cluster customers based on their engagement behaviors, distinguishing three behavioral profiles as moderate, low and high engagement shoppers. A decision matrix was formed where each unique combination of a customer's predicted intent and their given engagement behavior will determine a business's action. This decision matrix gives businesses a blueprint showing which customers they should send an advertisement in order to convert non-buyers into buyers, which type of customer would be best to upsell a product, or which customers to ignore, helping to maximize revenue while minimizing losses. This project highlights the importance of understanding your customers. Doing so will help increase the revenue of a given business. The value in understanding a customer's behavior not only lies in the revenue, but also will help improve the overall business.

Citations

Md Rayhan Kabir, Faisal Bin Ashraf, Rasif Ajwad, "Analysis of Different Predicting Model for Online Shoppers' Purchase Intention from Empirical Data", Semantic Scholar, <https://www.semanticscholar.org/paper/Analysis-of-Different-Predicting-Model-for-Online-Kabir-Ashraf/d82cb5bc3502fda14db68c6fa84ef1d3e3209675>

Online Shoppers Purchasing Intent Dataset, UC Irvine Machine Learning Repository, <https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>

Contributions

Team Member	Contribution
Peilin Li	Prediction Model, Exploratory visualization & analysis, Methods, Discussion
Maxwell Ho	Clustering Model, Introduction, Methods, Results, Discussion, Conclusion